

1. Consider the following districts in the map of Gujarat.



Color this map with the lowest number of colors posing it as a constraint satisfaction problem so that no two adjacent districts are colored the same. [Diu is not a district of Gujarat and Rann of Kutch and Little Rann of Kutch are also not districts]

2. Solve the following cryptarithmic problem posing it as a constraint satisfaction problem. Each letter represents a unique digit from 0 to 9. The leading letters (S and M) cannot be 0. Solve:

$$\begin{array}{r} \text{SEND} \\ + \text{MORE} \\ \hline \text{MONEY} \end{array}$$